VIHAAN X: SOFTWARE TRACK · DELHI TECHNOLOGICAL UNIVERSITY



DIGISCHEME

DIGITAL CITIZEN ASSISTANT

Multilingual access to government services & schemes — spoken in the language citizens actually think in.

Multiple scheduled languages. 1 assistant that speaks them all.

PRESENTED BY

iClaude X

TEAM MEMBERS

The people behind the assistant



Vinayak Mittal



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Mrinal Monga
(Team Leader)



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** Some PPTX features and interactive elements are not available in PDF format*

PROBLEM STATEMENT

TRACK DOMAIN:

OPEN INNOVATION

Digital Citizen Assistant for multilingual access to government services and schemes

WHAT THIS AIMS TO SOLVE

This aims to solve the full journey from unaware to actually benefiting — helping citizens discover what they qualify for, apply in their own language through verified sources, and complete the process themselves, with no gap left for middlemen to exploit.

WHY WE CHOSE THIS

We watched people close to us — students, families, workers — miss out on help they were entitled to, whether because they didn't know it existed or because they lost part of it to someone who "helped" them access it.

WHO FACES THE ISSUE?

- Students & families — missing scholarships/education schemes due to unclear eligibility
- Daily wage workers & farmers — unaware of subsidies, insurance, or pension schemes meant for them
- Elderly citizens — struggling with digital/language barriers to access healthcare or pension benefits
- Low-literacy or rural populations — most affected by complex forms and English-only portals
- Anyone without "system knowledge" — vulnerable to middlemen who exploit their lack of awareness

CURRENT GAP

HOW OUR SOLUTION FILLS THE GAP

WHO FACES THE ISSUE?

CURRENT GAP

- Scheme information is scattered across dozens of central/state portals
- Portals and forms are in English or formal Hindi
- The application process is manual and confusing — no guided help, unclear document requirements, high chance of rejection
- No verification layer — citizens can't tell official scheme sites from scam clones



HOW OUR SOLUTION FILLS THE GAP

WHO FACES THE ISSUE?

CURRENT GAP

HOW OUR SOLUTION FILLS THE GAP

- Discovery: One-time profile
- Language & literacy: Content in regional languages too
- Guided application: The assistant walks users through the actual form, auto-filling and generates a clear checklist of documents needed
- Trust: Verified against an official source list
- No middlemen needed: Because the process is guided end-to-end





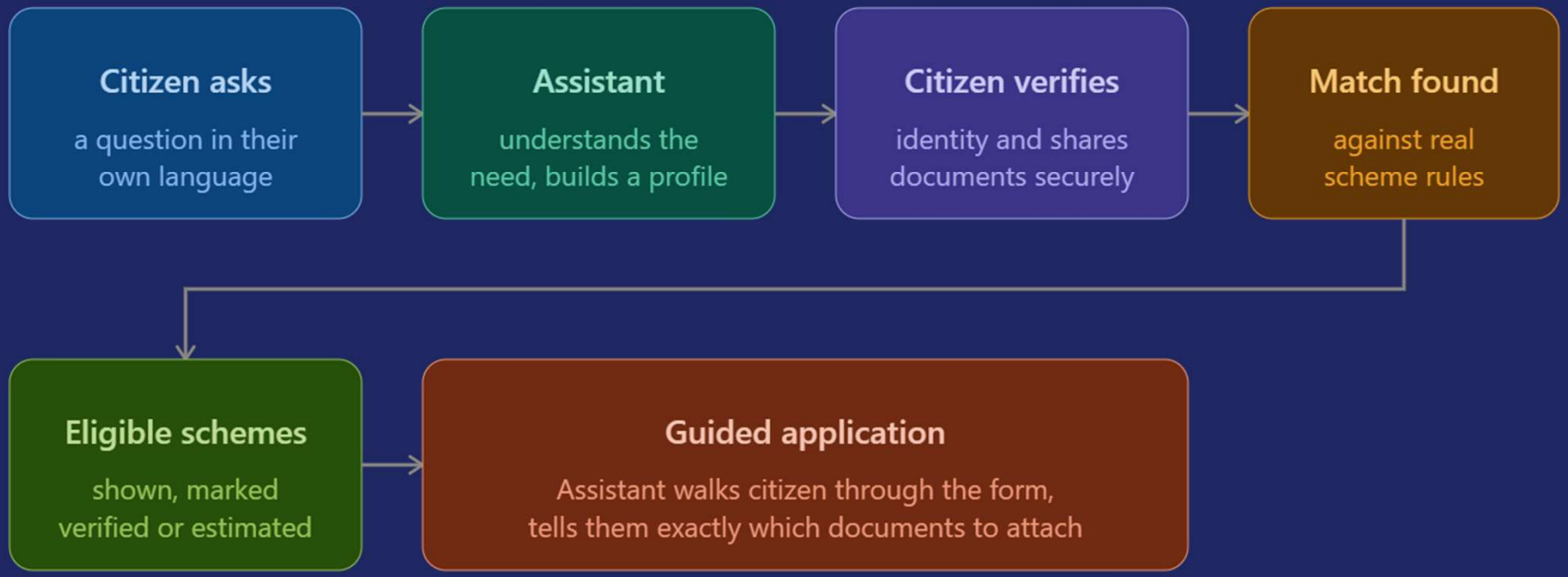
WHAT DOES THE REAL DATA SAY?

- "Studies show up to **1 in 4 eligible citizens** remain unaware of government schemes meant for them — and even among those who are aware, most don't know the **eligibility criteria** or **documents required** to apply.“

- "Scheme-impersonation scams aren't rare exceptions — cases span **PM-Kisan, Ayushman Bharat, Digital India, and Ujjwala Yojana**, with individual scams **cheating thousands of citizens** out of crores of rupees."

- "Middlemen take a cut of the **benefit over 50%** of the time in everyday citizen-government interactions — turning entitlements into a hidden transaction cost."

SOLUTION


Citizen asks
a question in their own language
Assistant
understands the need, builds a profile
Citizen verifies
identity and shares documents securely
Match found
against real scheme rules
Eligible schemes
shown, marked verified or estimated
Guided application
Assistant walks citizen through the form, tells them exactly which documents to attach

TECH STACK

Frontend / Access

- React.js (PWA) — web interface
- WhatsApp Business API (Twilio/Gupshup) — primary access channel, no app install required

Backend

- Node.js + Express.js — REST API server
- JWT — authentication and session handling
- MongoDB Atlas — database for user profiles and scheme data

Language & Voice

- Bhashini APIs — speech-to-text, text-to-speech, and translation across Indian languages

AI / Reasoning

- Gemini API — conversational assistant, form-filling guidance, plain-language explanations of eligibility

Verification & Trust

- DigiLocker Partner API (OAuth2) — pulls verified documents (income, caste, domicile certificates) with user consent
- Aadhaar eKYC (via licensed AUA/KUA) — identity verification
- .gov.in domain whitelist — protects against phishing/fake scheme websites

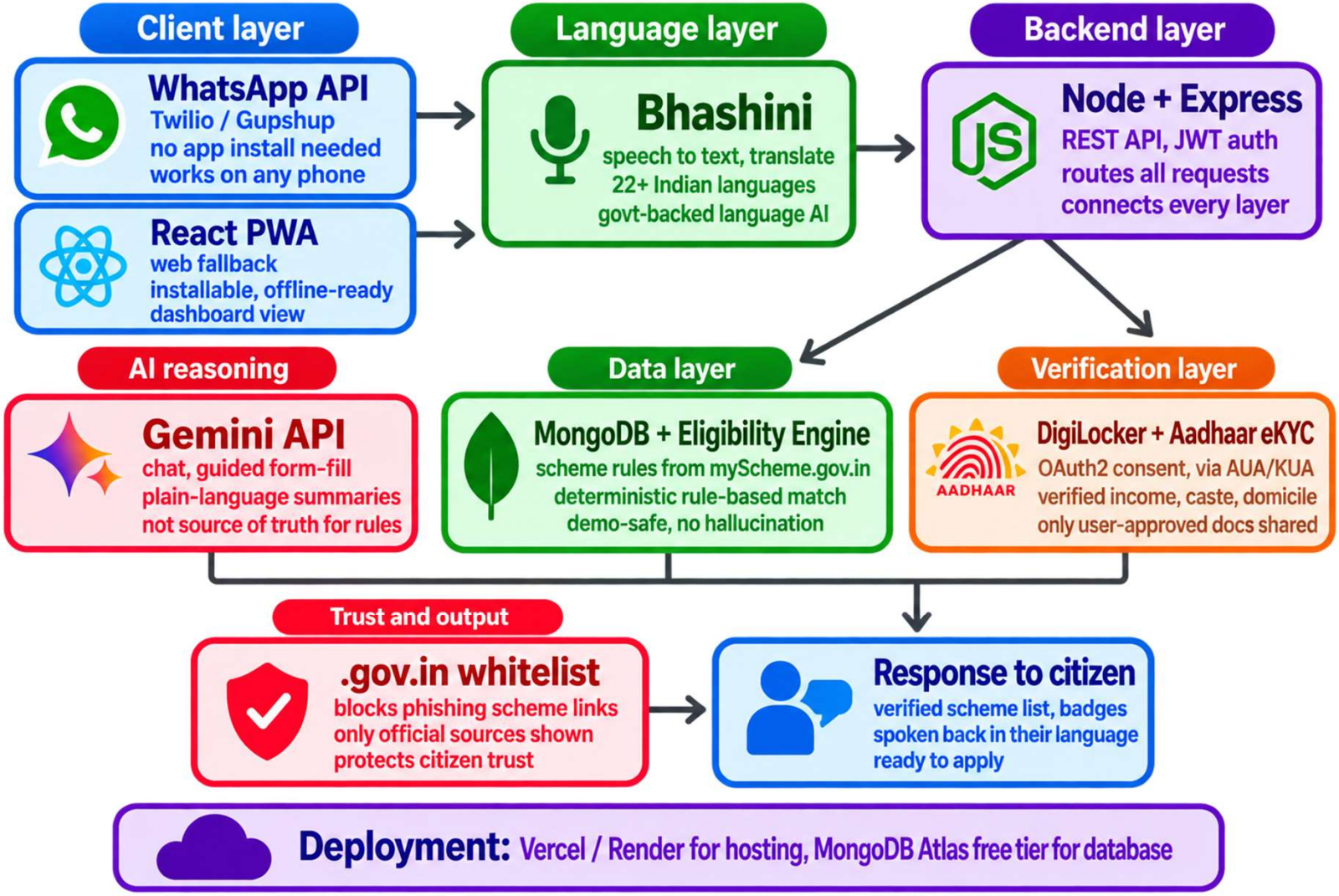
Data Source

- myScheme.gov.in — scheme eligibility rules and application links

Deployment

- Vercel / Render — hosting
- MongoDB Atlas — managed database (free tier for demo)

ARCHITECTURAL FLOW



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THANK YOU

Digital Citizen Assistant — every scheme, in every citizen's own language.

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